Advanced Models (Project Milestone 3, Group 001)

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Code and Data Access for the Instructor/Grader

Code and data for this project may be found on the project's GitHub page:

https://github.com/taydenkelleher/Stats-540-Regression-Project-Final-Data-Included

This .qmd file in particular is labeled as Advanced_Models.qmd, with the revised version matching the established naming standards being turned in on Canvas, as per project requirements.

The data for this project is found in the same GitHub repository under the file name final_data.csv.

Load packages

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.1
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(dplyr)
library(car)
```

```
Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode

The following object is masked from 'package:purrr':

    some
```

```
library(MASS)
```

```

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

    select
```

```
library(car)

# Including my traditional diagnostics package...  See the following for details:
# https://statacumen.com/teach/ADA1/help/Erhardt_R_Install.html
# The usual function call here is e_plot_lm_diagnostics(df, sw_plot_set = "simple")
if ((requireNamespace("erikmisc", quietly = TRUE)) == FALSE) {
  remotes::install_github("erikerhardt/erikmisc")
}
```

```
Warning: replacing previous import 'car::recode' by 'dplyr::recode' when
loading 'erikmisc'

Registered S3 methods overwritten by 'ggpp':
  method                  from
  heightDetails.titleGrob ggplot2
  widthDetails.titleGrob  ggplot2
```

```
Warning: replacing previous import 'ggplot2::stat_qq_line' by
'qqplotr::stat_qq_line' when loading 'erikmisc'

Warning: replacing previous import 'ggplot2::StatQqLine' by
'qqplotr::StatQqLine' when loading 'erikmisc'

Warning: replacing previous import 'dplyr::lag' by 'stats::lag' when loading
'erikmisc'

Warning: replacing previous import 'dplyr::filter' by 'stats::filter' when
loading 'erikmisc'

Warning: replacing previous import 'stats::filter' by 'dplyr::filter' when
loading 'erikmisc'

Warning: replacing previous import 'car::ellipse' by 'ellipse::ellipse' when
loading 'erikmisc'

Warning: replacing previous import 'ggplot2::annotate' by 'ggpp::annotate' when
loading 'erikmisc'
```

```r
library(erikmisc)

# Set seed for reproducability.
set.seed(6979)
```

Load the Data

```r
#Load the data
ed_data <- read.csv("final_data.csv")

#Remove entries without median debt reported, and
# turn it into numerical data.
ed_data <- ed_data |> subset(DEBT_MDN != "PS")
ed_data$DEBT_MDN <- as.numeric(ed_data$DEBT_MDN)

# Clean up all factors which were imported as integers.
ed_data$CONTROL.x <- as.factor(ed_data$CONTROL.x)
```

```r
ed_data$REGION <- as.factor(ed_data$REGION)
ed_data$HIGHDEG <-as.factor(ed_data$HIGHDEG)
ed_data$PREDDEG <- as.factor(ed_data$PREDDEG)
ed_data$NUMBRANCH <- as.factor(ed_data$NUMBRANCH)
ed_data$MAIN.x <-as.factor(ed_data$MAIN.x)
ed_data$SCH_DEG <- as.factor(ed_data$SCH_DEG)

# Turn all numeric data that got read as strings
# back into numerical data, changing NA values to
# zero for simplicity.
cols_to_convert <- c(
  "DEATH_YR8_RT",
  "COMP_ORIG_YR8_RT",
  "COMP_4YR_TRANS_YR8_RT",
  "PELL_COMP_4YR_TRANS_YR8_RT",
  "PELL_WDRAW_ORIG_YR8_RT",
  "NOPELL_WDRAW_ORIG_YR8_RT",
  "NOT1STGEN_COMP_4YR_TRANS_YR2_RT",
  "FIRSTGEN_COMP_ORIG_YR2_RT",
  "DEBT_MDN",
  "CREDDESC",
  "DEBT_ALL_STGP_ANY_MEAN",
  "BBRR1_FED_UG_DFLT"
)

new_ed_data <- ed_data %>%
  mutate(across(all_of(cols_to_convert), ~ {
    x <- suppressWarnings(as.numeric(.x))
    replace(x, is.na(x), 0)
  }))

# Now center our data. Since our original data had many character columns that should have be
centered_ed_data <- new_ed_data
numeric_cols <- sapply(centered_ed_data, is.numeric)
centered_ed_data[numeric_cols] <- lapply(centered_ed_data[numeric_cols],
                                         scale, scale = FALSE)
normalized_ed_data <- new_ed_data %>%
  mutate(across(where(is.numeric), ~as.vector(scale(.))))
rm(numeric_cols) #bruh why
```

1. Multiple regression models (10 pts)

At least two candidate models with several predictors are fit. Coefficients are interpreted correctly, holding the other predictors fixed. Categorical predictors are coded and interpreted correctly.

Model 1

Our first model is going to be the full model using the raw data for increased interpretability, minus four predictors, `CITY`, which we mostly just kept because it could be fun to look at the projected earnings later, and `CREDDESC`, which is entirely zeros, meaning we cannot use it. The last predictor we removed was `OPEID6`, which is the unique identifier for each college, which we wanted to remove so if colleges change over time, our models adjust more, rather than hardcoding the mean for each college. The last predictor we removed was `CONTROL.x`, which is exactly the same as `CONTROL.y` but coded as integers rather than the categorical variable. The last variable we got rid of was `TUITFTE` since it was causing infinite `vif`, due to it being identical to other `TUITFTE.x`. We still have 48 other predictors, so we are not lacking predictors.

Our model will look like this

$$Y_{ij} = \beta_0 + \sum_{i=1}^{48} \beta_j X_{ij} + \epsilon_{ij}$$

To interpret a few of the predictors, we have the beta value associated with `ADM_RATE_ALL` is `2.903e+03`, which means that if the normalized admission rate of the school you went to increases by 1, holding all else equal, then the median earnings of a student from that schools median income increases by `2.903e+03`. Note it is more applicable to say if the school acceptance rate of the school you attended increases by 1 percent (.01), then the predicted income of a graduate is `2.903`.

Another predictor to interpret is the beta value associated with `DEBT_MDN`, or the median original amount of the loan principal upon entering repayment of the school, which is `-1.460e-01`. Thus we have for every additional dollar of debt the median of your school, then the median earnings of the median earnings of graduates working and not enrolled 4 year after completing decreases by `.146`.

The last predictor we will interpret is the beta value associated with `CONTROL.y`, which is Control of institution, either public, private nonprofit, or private for profit. Since this is a categorical variable, it gets dummy coded. A beta value associated with `CONTROL.yPrivate, nonprofit` of `1.111e+03  5.100e+03` tells us that the median earnings of a graduates 4 years after graduating a private, nonprofit university is `1.111e+03` more than median earnings of graduates 4 years after graduating a Private for profit school. And a beta value associated with `CONTROL.yPublic` of `1.082e+03` tells us the median earnings of a graduates 4 years after

graduating a public university is 1.082e+03 more than median earnings of graduates 4 years after graduating a Private for profit school.

```
model_1 <- lm(EARN_MDN_4YR ~ .-CITY-CREDDESC-OPEID6-CONTROL.x-TUITFTE, data = new_ed_data)
summary(model_1)
```

```
Call:
lm(formula = EARN_MDN_4YR ~ . - CITY - CREDDESC - OPEID6 - CONTROL.x -
    TUITFTE, data = new_ed_data)

Residuals:
   Min     1Q Median     3Q    Max
-78705 -14775  -4478  10882 197697

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)
(Intercept)                   4.519e+03  2.380e+04   0.190 0.849411
X                            -6.466e-03  3.499e-02  -0.185 0.853401
SCH_DEG3                      6.944e+03  5.647e+03   1.230 0.218861
MAIN.x1                      -7.747e+03  1.385e+03  -5.595 2.22e-08 ***
NUMBRANCH2                   -6.612e+00  6.157e+02  -0.011 0.991431
NUMBRANCH3                    1.867e+02  8.824e+02   0.212 0.832412
NUMBRANCH4                    6.416e+03  2.279e+03   2.816 0.004869 **
NUMBRANCH5                    1.417e+03  8.241e+02   1.719 0.085574 .
NUMBRANCH6                   -6.303e+03  1.332e+03  -4.730 2.25e-06 ***
NUMBRANCH8                   -5.773e+03  1.892e+03  -3.051 0.002281 **
NUMBRANCH10                   2.938e+01  3.067e+03   0.010 0.992359
NUMBRANCH13                   1.753e+03  3.680e+03   0.476 0.633766
NUMBRANCH23                   6.966e+03  2.870e+03   2.427 0.015210 *
PREDDEG3                     -7.202e+03  5.747e+03  -1.253 0.210146
HIGHDEG3                      6.105e+03  2.294e+04   0.266 0.790156
HIGHDEG4                      9.455e+03  2.294e+04   0.412 0.680289
ST_FIPS                       1.772e+01  1.998e+01   0.887 0.375214
REGION2                       4.848e+02  6.504e+02   0.745 0.456064
REGION3                       8.983e+01  7.215e+02   0.125 0.900913
REGION4                      -1.010e+03  8.311e+02  -1.216 0.224110
REGION5                      -1.445e+03  7.436e+02  -1.943 0.052013 .
REGION6                      -6.153e+02  9.304e+02  -0.661 0.508387
REGION7                       4.870e+02  1.022e+03   0.476 0.633735
REGION8                       1.794e+03  7.768e+02   2.309 0.020957 *
REGION9                       3.124e+03  1.621e+04   0.193 0.847147
```

```
ADM_RATE_ALL                       2.903e+03  1.116e+03   2.601 0.009297 **
PCIP12                            -1.619e+05  5.482e+04  -2.954 0.003138 **
PCIP24                            -3.985e+03  2.973e+03  -1.341 0.180067
PCIP51                             1.699e+04  1.428e+03  11.901  < 2e-16 ***
PCIP52                             1.103e+04  1.874e+03   5.888 3.95e-09 ***
UGDS                               1.548e-01  7.003e-02   2.210 0.027090 *
UG25ABV                           -1.130e+03  3.775e+03  -0.299 0.764699
UGDS_WHITE                        -6.837e+03  1.997e+03  -3.424 0.000619 ***
UGDS_BLACK                         7.695e+01  2.494e+03   0.031 0.975384
UGDS_HISP                         -5.923e+03  2.411e+03  -2.457 0.014030 *
PPTUG_EF                           1.117e+04  3.718e+03   3.004 0.002666 **
COSTT4_A                           7.565e-02  5.858e-02   1.291 0.196575
TUITIONFEE_IN                     -6.045e-02  7.426e-02  -0.814 0.415604
TUITIONFEE_OUT                     4.464e-02  3.646e-02   1.224 0.220856
INEXPFTE                           1.827e-01  2.220e-02   8.232  < 2e-16 ***
AVGFACSAL                          1.514e+00  1.229e-01  12.326  < 2e-16 ***
PFTFAC                             4.938e+02  8.227e+02   0.600 0.548349
C150_4                            -2.470e+04  4.947e+03  -4.992 6.01e-07 ***
PFTFTUG1_EF                        3.930e+03  2.185e+03   1.798 0.072148 .
D150_4                            -7.743e-01  3.038e-01  -2.549 0.010819 *
C150_4_WHITE                       1.002e+04  3.568e+03   2.809 0.004973 **
C150_4_BLACK                       1.044e+03  1.495e+03   0.698 0.485085
C150_4_HISP                        1.550e+03  1.629e+03   0.951 0.341593
DEATH_YR8_RT                       1.782e+05  8.246e+04   2.161 0.030740 *
COMP_ORIG_YR8_RT                   1.839e+04  3.560e+03   5.167 2.40e-07 ***
COMP_4YR_TRANS_YR8_RT             -1.778e+04  1.074e+04  -1.655 0.097847 .
PELL_COMP_4YR_TRANS_YR8_RT         2.251e+04  9.822e+03   2.292 0.021920 *
PELL_WDRAW_ORIG_YR8_RT            -1.238e+04  5.860e+03  -2.113 0.034632 *
NOPELL_WDRAW_ORIG_YR8_RT           2.571e+04  6.405e+03   4.014 5.98e-05 ***
NOT1STGEN_COMP_4YR_TRANS_YR2_RT    4.682e+04  1.166e+04   4.016 5.94e-05 ***
FIRSTGEN_COMP_ORIG_YR2_RT         -2.616e+03  2.893e+03  -0.904 0.365763
TUITFTE.1                          5.430e-02  4.423e-02   1.228 0.219587
OVERALL_YR2_N                     -2.947e-01  2.082e-01  -1.415 0.156944
DEBT_MDN                          -1.460e-01  6.387e-02  -2.286 0.022256 *
DEBT_ALL_STGP_ANY_MEAN            -3.954e-01  1.134e-02 -34.877  < 2e-16 ***
BBRR1_FED_UG_DFLT                 -3.317e+05  7.080e+04  -4.684 2.82e-06 ***
DISTANCE                           2.398e+03  2.186e+02  10.969  < 2e-16 ***
CONTROL.yPrivate, nonprofit        1.111e+03  5.100e+03   0.218 0.827596
CONTROL.yPublic                    1.082e+03  5.199e+03   0.208 0.835214
SATVRMID                          -5.887e+01  1.189e+01  -4.950 7.48e-07 ***
SATMTMID                           1.064e+02  9.814e+00  10.839  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 22740 on 30667 degrees of freedom
Multiple R-squared:  0.2051,    Adjusted R-squared:  0.2034
F-statistic: 121.7 on 65 and 30667 DF,  p-value: < 2.2e-16
```

Model 2

Our next model is a more refined model, only using the statistically significant terms, along with statistically significant interaction terms.

Polynomial Terms

The way we chose these terms is we would fit models and try different interaction and polynomial terms, and if there statistically significant predictors, we would keep them, and if not, we would remove them and try something else. For terms with very small p-values, which tell us given $\beta = 0$, what is the probability that our data is at least this spread out, is very small, meaning the null model is likely wrong and there is some relationship, we chose to fit some polynomial terms, including `C150_4_WHITE` and `UGDS_WHITE`, both degree 2 to help with interpretability and because higher order polynomial terms were not as statistically significant.

`C150_4_WHITE` is Completion rate for first-time, full-time students at four-year institutions (150% of expected time to completion) for white students. Completing college is likely a good predictor of earnings after college, so it makes sense this term is statistically significant, and it could be nonlinear. We had 4 versions of the `C150_4` terms, and chose the completion rate for white students because it had the smallest p value in the full model.

We did a similar thing for `UGDS_WHITE`, which is the proportion of students that are white. We suspected that the data was nonlinear, and wanted to investigate it as a polynomial.

Interaction Terms

We also looked at a few interaction terms. The `PCIP51` and `PCIP52` terms are "Percentage of degrees awarded in Health Professions And Related Programs" and "Percentage of degrees awarded in Business, Management, Marketing, And Related Support Services." Our "full" model had 4 of these terms, out of the 52 our data-set had. The two we chose had the most statistical significance, but we suspected that they interacted since other degrees exist, and degree type likely impacts earnings, so we are measuring other degrees.

Another group of interaction terms we got was `AVGFACSAL`, `DEBT_ALL_STGP_ANY_MEAN`, and `DEBT_MDN`, which are average faculty salary, Average Stafford and Grad PLUS loan debt disbursed at all institutions, and The median original amount of the loan principal upon entering repayment. All of these terms were statistically significant, and we were curious if they interacted, and if so how much they interacted, since staff salary may correlate with increased

costs and higher debt. We were also curious how mean debt interacted with median debt, since mean is more impacted by heavy tails and we wanted some way to measure distribution of student debt.

This gives us the following models:

$$\begin{aligned}\hat{Y} = {} & 4.940 \times 10^3 - 6.271 \times 10^3 X_1 + 4.897 \times 10^3 X_2 \\ & - 1.674 \times 10^5 X_3 - 7.406 \times 10^4 X_4 \\ & + 1.910 \times 10^4 X_5 + 1.449 \times 10^4 X_6 \\ & - 2.592 \times 10^5 X_7 - 1.404 \times 10^5 X_8 \\ & + 1.240 \times 10^{-1} X_9 + 1.603 X_{10} \\ & - 4.528 \times 10^{-1} X_{11} + 5.050 \times 10^{-2} X_{12} \\ & + 2.316 \times 10^4 X_{13} + 4.876 \times 10^4 X_{14} \\ & + 2.988 \times 10^4 X_{15} + 2.031 \times 10^4 X_{16} \\ & + 2.130 \times 10^3 X_{17} - 6.292 \times 10^1 X_{18} \\ & + 9.834 \times 10^1 X_{19} \\ & + 4.034 \times 10^4 X_5 X_6 \\ & - 9.998 \times 10^{-5} X_{10} X_{11} \\ & - 8.150 \times 10^{-5} X_{10} X_{12} \\ & + 2.350 \times 10^{-5} X_{11} X_{12} \\ & + 6.686 \times 10^6 X_{14} X_{15} \\ & + 1.510 \times 10^{-1} X_{18} X_{19}\end{aligned}$$

Interpretation

To interpret what a few of these values mean, we have the beta value associate with `ADM_RATE_ALL` is `4.897e+03`. Since this is not a part of any interaction terms or polynomial terms, we know that if the admission rate of the college increased by 1 percent, then the predicted Median earnings of graduates working and not enrolled 4 year after completing would increase `48.97`.

The beta value associated with `DEBT_MDN` is `-1.891e-01`, which means that the linear component of the `DEBT_MDN` is `-1.891e-01`.

The beta value associated with `AVGFACSAL:DEBT_MDN`, the interaction between these two predictors is `-8.150e-05` which means the slope of `DEBT_MDN` increases by `-8.150e-05` for every increase of `AVGFACSAL` by 1, and vice versa. The beta value associated with `DEBT_ALL_STGP_ANY_MEAN:DEBT_MDN` is `2.350e-05`, which means that the slope of

DEBT_ALL_STGP_ANY_MEAN increases by 2.350e-05 for every increase of 1 in DEBT_MDN, and vice versa.

We did not have any categorical variables in this to interpret for this model.

```
model_2 <- lm(EARN_MDN_4YR~MAIN.x  + ADM_RATE_ALL+poly(C150_4_WHITE,2) + (PCIP51 + PCIP52)^2
summary(model_2)
```

```
Call:
lm(formula = EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE,
    2) + (PCIP51 + PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE +
    (AVGFACSAL + DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2, data = centered_ed_data)

Residuals:
   Min     1Q Median     3Q    Max
-80674 -14136  -4721  10695 195791

Coefficients:
                                        Estimate Std. Error t value Pr(>|t|)
(Intercept)                            4.940e+03  5.599e+02   8.823  < 2e-
16
MAIN.x1                               -6.271e+03  5.925e+02 -10.583  < 2e-
16
ADM_RATE_ALL                           4.897e+03  1.047e+03   4.676 2.94e-
06
poly(C150_4_WHITE, 2)1                -1.674e+05  5.909e+04  -2.832 0.004623
poly(C150_4_WHITE, 2)2                -7.406e+04  3.443e+04  -2.151 0.031461
PCIP51                                 1.910e+04  1.485e+03  12.860  < 2e-
16
PCIP52                                 1.449e+04  1.721e+03   8.424  < 2e-
16
poly(UGDS_WHITE, 2)1                  -2.592e+05  2.944e+04  -8.805  < 2e-
16
poly(UGDS_WHITE, 2)2                  -1.404e+05  2.543e+04  -5.521 3.39e-
08
INEXPFTE                               1.240e-01  2.105e-02   5.890 3.90e-
09
AVGFACSAL                              1.603e+00  9.071e-02  17.667  < 2e-
16
```

```
DEBT_ALL_STGP_ANY_MEAN                  -4.528e-01  1.132e-02 -40.012  < 2e-
16
DEBT_MDN                                 5.050e-02  4.826e-02   1.046 0.295355
COMP_ORIG_YR8_RT                         2.316e+04  2.500e+03   9.263  < 2e-
16
DEATH_YR8_RT                             4.876e+04  6.993e+04   0.697 0.485633
PELL_COMP_4YR_TRANS_YR8_RT               2.988e+04  5.302e+03   5.636 1.75e-
08
NOPELL_WDRAW_ORIG_YR8_RT                 2.031e+04  4.033e+03   5.035 4.81e-
07
DISTANCE                                 2.130e+03  2.125e+02  10.024  < 2e-
16
SATVRMID                                -6.292e+01  9.947e+00  -6.325 2.56e-
10
SATMTMID                                 9.834e+01  8.377e+00  11.739  < 2e-
16
PCIP51:PCIP52                            4.034e+04  1.155e+04   3.492 0.000481
AVGFACSAL:DEBT_ALL_STGP_ANY_MEAN        -9.998e-05  4.497e-06 -22.232  < 2e-
16
AVGFACSAL:DEBT_MDN                      -8.150e-05  1.549e-05  -5.262 1.44e-
07
DEBT_ALL_STGP_ANY_MEAN:DEBT_MDN          2.350e-05  3.510e-06   6.694 2.20e-
11
DEATH_YR8_RT:PELL_COMP_4YR_TRANS_YR8_RT  6.686e+06  2.011e+06   3.325 0.000884
SATVRMID:SATMTMID                        1.510e-01  4.009e-02   3.766 0.000166

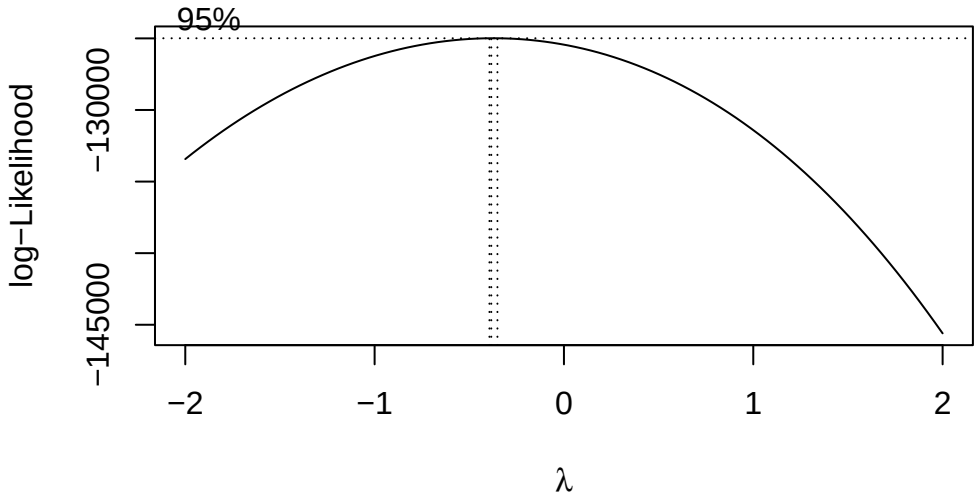
(Intercept)                             ***
MAIN.x1                                 ***
ADM_RATE_ALL                            ***
poly(C150_4_WHITE, 2)1                  **
poly(C150_4_WHITE, 2)2                  *
PCIP51                                  ***
PCIP52                                  ***
poly(UGDS_WHITE, 2)1                    ***
poly(UGDS_WHITE, 2)2                    ***
INEXPFTE                                ***
AVGFACSAL                               ***
DEBT_ALL_STGP_ANY_MEAN                  ***
DEBT_MDN
COMP_ORIG_YR8_RT                        ***
DEATH_YR8_RT
PELL_COMP_4YR_TRANS_YR8_RT              ***
NOPELL_WDRAW_ORIG_YR8_RT                ***
```

```
DISTANCE                                    ***
SATVRMID                                    ***
SATMTMID                                    ***
PCIP51:PCIP52                               ***
AVGFACSAL:DEBT_ALL_STGP_ANY_MEAN            ***
AVGFACSAL:DEBT_MDN                          ***
DEBT_ALL_STGP_ANY_MEAN:DEBT_MDN             ***
DEATH_YR8_RT:PELL_COMP_4YR_TRANS_YR8_RT     ***
SATVRMID:SATMTMID                           ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 22600 on 30707 degrees of freedom
Multiple R-squared:  0.2139,    Adjusted R-squared:  0.2133
F-statistic: 334.3 on 25 and 30707 DF,  p-value: < 2.2e-16
```

Model 3

```r
# Find the lm for the unshifted data, so we can get a boxcox model.
model_3_unshifted <- lm(
  EARN_MDN_4YR ~ (TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL)^2
  + PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN,
  data = ed_data
)
# Generate boxcox model, and show plot.  Then, transform data.
bc3 <- boxcox(model_3_unshifted)
```




```
#bc3
bc3$x[which.max(bc3$y)]
```

```
[1] -0.3838384
```

```
m3_transformed_data = ed_data
# Apply boxcox transform.
m3_transformed_data$EARN_MDN_4YR = ((ed_data$EARN_MDN_4YR^bc3$x[which.max(bc3$y)] - 1)/bc3$x
# Do some cleaning up.
rm(model_3_unshifted)
rm(bc3)
# Now center the data.
centered_m3_data <- m3_transformed_data
numeric_cols <- sapply(m3_transformed_data, is.numeric)
centered_m3_data[numeric_cols] <- lapply(centered_m3_data[numeric_cols],
                                         scale, scale = FALSE)
rm(numeric_cols)
# And finally, generate the actual model and give the summary.
model_3 <- lm(
  EARN_MDN_4YR ~ (TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL)^2
  + PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN,
  data = centered_m3_data
```

```
)
summary(model_3)
```

```
Call:
lm(formula = EARN_MDN_4YR ~ (TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL)^2 +
    PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN,
    data = centered_m3_data)

Residuals:
      Min        1Q    Median        3Q       Max
-0.027375 -0.003334 -0.000345  0.003481  0.018811

Coefficients:
                             Estimate Std. Error t value Pr(>|t|)
(Intercept)                -7.157e-03  4.941e-03  -1.448   0.1475
TUITIONFEE_IN               2.425e-07  1.301e-08  18.644  < 2e-16 ***
CONTROL.x2                 -4.087e-03  1.763e-04 -23.187  < 2e-16 ***
CONTROL.x3                  7.901e-04  2.457e-03   0.322   0.7478
ADM_RATE_ALL               -4.019e-03  3.473e-04 -11.574  < 2e-16 ***
PREDDEG3                    1.016e-03  4.146e-04   2.451   0.0143 *
HIGHDEG3                    7.124e-03  4.957e-03   1.437   0.1507
HIGHDEG4                    8.216e-03  4.957e-03   1.658   0.0974 .
C150_4                      3.905e-03  2.583e-04  15.119  < 2e-16 ***
DEBT_MDN                   -2.851e-08  1.124e-08  -2.536   0.0112 *
TUITIONFEE_IN:CONTROL.x2   -1.368e-07  1.292e-08 -10.590  < 2e-16 ***
TUITIONFEE_IN:CONTROL.x3   -5.906e-07  1.306e-07  -4.523 6.11e-06 ***
TUITIONFEE_IN:ADM_RATE_ALL -7.555e-08  2.666e-08  -2.834   0.0046 **
CONTROL.x2:ADM_RATE_ALL     5.558e-03  9.440e-04   5.888 3.95e-09 ***
CONTROL.x3:ADM_RATE_ALL     5.190e-03  1.568e-02   0.331   0.7407
ADM_RATE_ALL:C150_4        -6.031e-03  1.204e-03  -5.008 5.54e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.004938 on 30717 degrees of freedom
Multiple R-squared:  0.1153,    Adjusted R-squared:  0.1148
F-statistic: 266.8 on 15 and 30717 DF,  p-value: < 2.2e-16
```

For our third, model, we took a more hand-built approach. We included in-state tuition fee, institution type (public, private non-profit, or private for-profit), and admission rate, as well as all degree two interaction terms between these; predicted degree accomplishment, highest

degree achieved, completion rate, median debt, and the interaction term between completion rate and institution admission rate.

Preemptively, we examine the Box-Cox plot, and find that a transform of the data is needed. This is backed up by the QQ-plot not being very linear, and the histogram of residuals not following a normal distribution, although we do not include those here as they do not directly relate to the final model. After this transform, we center our data to eliminate non-essential contributions of interaction and polynomial terms with the base variables. Here, we do not pick a convenient value inside our confidence interval, as our confidence interval is so narrow that there are no particularly nice values inside of it.

After this transform, we label our variables as follows: Y as our transformed and centered version of four-year median earnings, X_1 as our centered in-state tuition, X_2 as our indicator for being a private nonprofit institution, X_3 as our indicator for being a private for-profit institution, X_4 as our centered admission rate as a decimal, X_5 as our indicator for predicted degree, X_6 and X_7 as our indicators for the various options of highest degree offered (master's and doctoral), X_8 as our centered graduation rate, and X_9 as our centered median debt. This gives us the following model:

$$Y = 10^{-3}(-7.157+2.425\cdot10^{-4}X_1-4.087X_2+0.7901X_3-4.019X_4+1.016X_5+7.124X_6+8.216X_7+3.905X_8-2.851\cdot$$

Here, we factor the 10^{-3} term out to save space.

To interpret this model, we begin by noting that in this model, the values of Y are rather small, due both to the Box-Cox transform and the centering. As such, we go in with the knowledge that each variable is, itself, going to be rather small.

For the intercept, $-7.157 \cdot 10^{-3}$, we know that this is the contribution to Y in the case that a school is numerically average, with each of our indicators as zero. This is notably not the same as being an average university. The indicated p-value shows that this is likely zero.

For the coefficient of X_1, the in-state tuition, is $2.425 \cdot 10^{-7}$. The smaller coefficient here makes sense, as in-state tuition tend to be measured in thousands of dollars. The interpretation here is that the direct (linear) contribution of in-state tuition is approximately $2.425 \cdot 10^{-7}$, meaning that the higher the in-state tuition of a university, the higher your median income four years after graduation will be. The p-value of this coefficient is low, indicating that it is very likely non-zero.

For the coefficient of X_2, we find that for private nonprofit institutions, the transformed and centered median income decreases by $4.087 \cdot 10^{-3}$. The p-value for this coefficient is also low, similar to the above, indicating that it is likely non-zero.

For the coefficient of X_3, we find that for private for-profit institutions, the expected contribution to the transformed and adjusted median income increases by $7.901 \cdot 10^{-4}$. From the associated t-test and p-value, we know that this is very likely to actually be zero.

For the coefficient of X_4, we find that for every 1 your decimal expression of acceptance rate increases from the mean national acceptance rate, the transformed and shifted four-year median income after graduation goes down by $4.019 \cdot 10^{-3}$. For a more readable interpretation, the more selective an institution, the higher the median pay is likely to be four years after graduation. The associated t-test tells us that this effect is significant, and that the coefficient is almost certainly not zero.

Similar interpretations are available for the remaining variables. However, following the discussion after class with the instructor, they are omitted here due to our already excessive number of variables, so as to keep this write-up to a readable length.

2. Interactions and polynomial terms (10 pts)

At least one interaction or polynomial term is considered, with a reason for it. It is tested (for example, a partial F test or a test) and the result is stated. An interaction plot or fitted-curve plot is included where relevant. A well-reasoned term that turns out not to help still earns full credit.

Model 1

There are no interaction terms with this model. The full model is just meant to compare and interpret everything against each other. Therefore there is no interaction terms to allow for easy interpret ability. Additional interaction terms would unnecessarily complicate the model.

Model 2

To test if the polynomial and interaction terms are significant, we will train reduced models that have everything except the polynomial or interaction term. Then we will use the `anova` function to compute a partial F test. After doing that, we see that each term is significant.

Model 2

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C150_WHITE, Degree 2 Polynomial Term

One variable that had a small p value in the full model was C150_WHITE, which is Completion rate for first-time, full-time students at four-year institutions (150% of expected time to completion) for white students. There were some other C150 terms, including pooled term, and categories for black and hispanic students. These other terms were not super significant, so were looked to drop some of the variables, but wanted to keep some nonlinear information from the variable, so we decided to add a degree 2 polynomial term. After doing a partial f test, we have a p value of 0.000417, which suggests the linear model does not explain everything.

```
model_2_reduced <- lm(EARN_MDN_4YR~MAIN.x  + ADM_RATE_ALL+(C150_4_WHITE) + (PCIP51 + PCIP52)
print(anova(model_2_reduced, model_2))
```

```
Analysis of Variance Table

Model 1: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + (C150_4_WHITE) + (PCIP51 +
    PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT) + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
Model 2: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
  Res.Df        RSS Df  Sum of Sq      F   Pr(>F)
1  30709 1.5688e+13
2  30707 1.5680e+13  2 7950078109 7.7844 0.000417 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
rm(model_2_reduced)
```

UGDS_WHITE Variable, Degree 2 Polynomial Term

UGDS_WHITE, which represents Total share of enrollment of undergraduate degree-seeking students who are white, is similar to C150_WHITE. There are other UGDS terms, the nominal UGDS and specific ones for black and hispanic students, but again the UGDS_WHITE had the smallest p value in the model, and we wanted to drop variables, but not lose nonlinear data, so we added a polynomial term. After running a partial f test, we get that the p value is 3.394e-08 which suggests the polynomial terms explains real variation in the data.

```
model_2_reduced <- lm(EARN_MDN_4YR~MAIN.x  + ADM_RATE_ALL+poly(C150_4_WHITE,2) + (PCIP51 + PC
print(anova(model_2_reduced, model_2))
```

```
Analysis of Variance Table

Model 1: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52)^2 + UGDS_WHITE + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
Model 2: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
  Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
1  30708 1.5696e+13
2  30707 1.5680e+13  1 1.5566e+10 30.484 3.394e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
rm(model_2_reduced)
```

Interaction of PCIP51 and PCIP52

For these terms, they represent Percentage of degrees awarded in Health Professions And Related Programs and Percentage of degrees awarded in Business, Management, Marketing, And Related Support Services respectively. Again, we have 50 other terms of this style representing share of degrees awarded by other departments. We wanted this to be interpretable, but to include some of the nonlinear information in the data. After computing a partial f test, we get a p-value of `0.0004809`, which suggests our interaction term represents a real interaction in the data.

```
model_2_reduced <- lm(EARN_MDN_4YR~MAIN.x  + ADM_RATE_ALL+poly(C150_4_WHITE,2) + (PCIP51 + PC
print(anova(model_2_reduced, model_2))
```

```
Analysis of Variance Table

Model 1: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52) + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
```

```
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
Model 2: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
  Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
1  30708 1.5687e+13
2  30707 1.5680e+13  1 6225303993 12.191 0.0004809 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
rm(model_2_reduced)
```

Interaction of AVGFACSAL, DEBT_ALL_STGP_ANY_MEAN, and DEBT_MDN

AVGFACSAL is average faculty salary, DEBT_ALL_STGP_ANY_MEAN is Average Stafford and Grad PLUS loan debt disbursed at all institutions, and The median original amount of the loan principal upon entering repayment at the institution. All of these likely correlate to higher costs of schooling which could impact career earnigns, and we wanted to see how they interact with each other. Computing a partial f test, we get a p value of `2.2e-16` which suggests our interaction has real effects on the data.

```
model_2_reduced <- lm(EARN_MDN_4YR~MAIN.x  + ADM_RATE_ALL+poly(C150_4_WHITE,2) + (PCIP51 + PC
print(anova(model_2_reduced, model_2))
```

```
Analysis of Variance Table

Model 1: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN) + COMP_ORIG_YR8_RT + (DEATH_YR8_RT +
    PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
Model 2: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
  Res.Df       RSS Df  Sum of Sq      F    Pr(>F)
```

```
1  30710 1.596e+13
2  30707 1.568e+13  3 2.7997e+11 182.76 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
rm(model_2_reduced)
```

Interaction of DEATH_YR8_RT and PELL_COMP_4YR_TRANS_YR8_RT

DEATH_YR8_RT is Percent died within 8 years at original institution and PELL_COMP_4YR_TRANS_YR8 represents Percent of students who received a Pell Grant at the institution and who transferred to a 4-year institution and completed within 8 years. Death rate likely correlates with other things that correlate to worse education, and being elegible for Pell Grant also likely correlates with these confounders that make completing education more difficult. We were curious if these had any real interaction. After completing a partial f test, we got a p value of `0.000884` which suggests the interaction is statistically significant.

```
model_2_reduced <- lm(EARN_MDN_4YR~MAIN.x  + ADM_RATE_ALL+poly(C150_4_WHITE,2) + (PCIP51 + PC
print(anova(model_2_reduced, model_2))
```

```
Analysis of Variance Table

Model 1: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT) + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
Model 2: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
  Res.Df        RSS Df  Sum of Sq      F   Pr(>F)
1  30708 1.5686e+13
2  30707 1.5680e+13  1 5646761422 11.058 0.000884 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
rm(model_2_reduced)
```

Interaction of SATVRMID and SATMTMID

The last interaction term we used was between the two SAT scores. In EDA, we observed these two variables are strongly correlated with each other, but in the full model they were strongly correlated with each other. Thus we wanted to do see what the interaction was and what would happen. After running a partial f test, we get that a p value of `0.0001664` which suggest the interaction is real. One other potential explaination is that this is bascially a squared term for SAT scores, since the SAT scores are so strongly correlated and SAT scores predict earning of graduates.

```
model_2_reduced <- lm(EARN_MDN_4YR~MAIN.x  + ADM_RATE_ALL+poly(C150_4_WHITE,2) + (PCIP51 + PC
print(anova(model_2_reduced, model_2))
```

```
Analysis of Variance Table

Model 1: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)
Model 2: EARN_MDN_4YR ~ MAIN.x + ADM_RATE_ALL + poly(C150_4_WHITE, 2) +
    (PCIP51 + PCIP52)^2 + poly(UGDS_WHITE, 2) + INEXPFTE + (AVGFACSAL +
    DEBT_ALL_STGP_ANY_MEAN + DEBT_MDN)^2 + COMP_ORIG_YR8_RT +
    (DEATH_YR8_RT + PELL_COMP_4YR_TRANS_YR8_RT)^2 + NOPELL_WDRAW_ORIG_YR8_RT +
    DISTANCE + (SATVRMID + SATMTMID)^2
  Res.Df        RSS Df  Sum of Sq     F    Pr(>F)
1  30708 1.5688e+13
2  30707 1.5680e+13  1 7240941571 14.18 0.0001664 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
rm(model_2_reduced)
```

Model 3

Here, we include degree two interaction terms between in-state tuition, institution type, and admission rate. This is because the effect of in-state tuition can be mitigated by private institutions waiving tuition, in-state tuition can often be waived by higher admittance requirements (reflected by admission rate), and institution type and admission rate together can tell us how selective the institution is, often indicating how well perceived the institution is in the job market.

There is also room for C150_4 to have interaction terms with practically every other variable present, as drop-out rate is a very useful indicator of median pay after graduation. Given the nature of graduation rate and admittance criteria, they will likely have interaction in that the more strict the admittance criteria, the higher the graduation rate will likely be. As such, we try to account for how this effects pay by including an interaction term.

The remaining items: median debt, predicted degree, and highest degree received, all indicate base-line requirements for most students at these institutions, and help us account for what the average student at these institutions look like.

Similar ANOVA models to those found above are included below.

Interaction of TUITIONFEE_IN and CONTROL.x

```
model_3_reduced <- lm(
  EARN_MDN_4YR ~ TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL +
  TUITIONFEE_IN:ADM_RATE_ALL + CONTROL.x:ADM_RATE_ALL +
  PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN,
  data = centered_m3_data
)
anova(model_3_reduced, model_3)
```

```
Analysis of Variance Table

Model 1: EARN_MDN_4YR ~ TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL + TUITIONFEE_IN:ADM_RATE_ALL
    CONTROL.x:ADM_RATE_ALL + PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL +
    DEBT_MDN
Model 2: EARN_MDN_4YR ~ (TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL)^2 +
    PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN
  Res.Df     RSS Df Sum of Sq      F    Pr(>F)
1  30719 0.75205
2  30717 0.74896  2 0.0030872 63.307 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
rm(model_3_reduced)
```

Interaction of TUITIONFEE_IN and ADM_RATE_ALL

```r
model_3_reduced <- lm(
  EARN_MDN_4YR ~ TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL +
  TUITIONFEE_IN:CONTROL.x + CONTROL.x:ADM_RATE_ALL +
  PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN,
  data = centered_m3_data
)
anova(model_3_reduced, model_3)
```

```
Analysis of Variance Table

Model 1: EARN_MDN_4YR ~ TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL + TUITIONFEE_IN:CONTROL.x +
    CONTROL.x:ADM_RATE_ALL + PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL +
    DEBT_MDN
Model 2: EARN_MDN_4YR ~ (TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL)^2 +
    PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN
  Res.Df     RSS Df  Sum of Sq      F   Pr(>F)
1  30718 0.74916
2  30717 0.74896  1 0.00019579 8.0298 0.004604 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
rm(model_3_reduced)
```

Interaction of CONTROL.x and ADM_RATE_ALL

```r
model_3_reduced <- lm(
  EARN_MDN_4YR ~ TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL +
  TUITIONFEE_IN:CONTROL.x + TUITIONFEE_IN:ADM_RATE_ALL +
  PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN,
  data = centered_m3_data
)
anova(model_3_reduced, model_3)
```

```
Analysis of Variance Table

Model 1: EARN_MDN_4YR ~ TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL + TUITIONFEE_IN:CONTROL.x +
    TUITIONFEE_IN:ADM_RATE_ALL + PREDDEG + HIGHDEG + C150_4 +
    C150_4:ADM_RATE_ALL + DEBT_MDN
Model 2: EARN_MDN_4YR ~ (TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL)^2 +
    PREDDEG + HIGHDEG + C150_4 + C150_4:ADM_RATE_ALL + DEBT_MDN
```

```
  Res.Df     RSS Df  Sum of Sq      F    Pr(>F)
1  30719 0.74981
2  30717 0.74896  2 0.00084576 17.343 2.966e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
rm(model_3_reduced)
```

3. Diagnostics (10 pts)

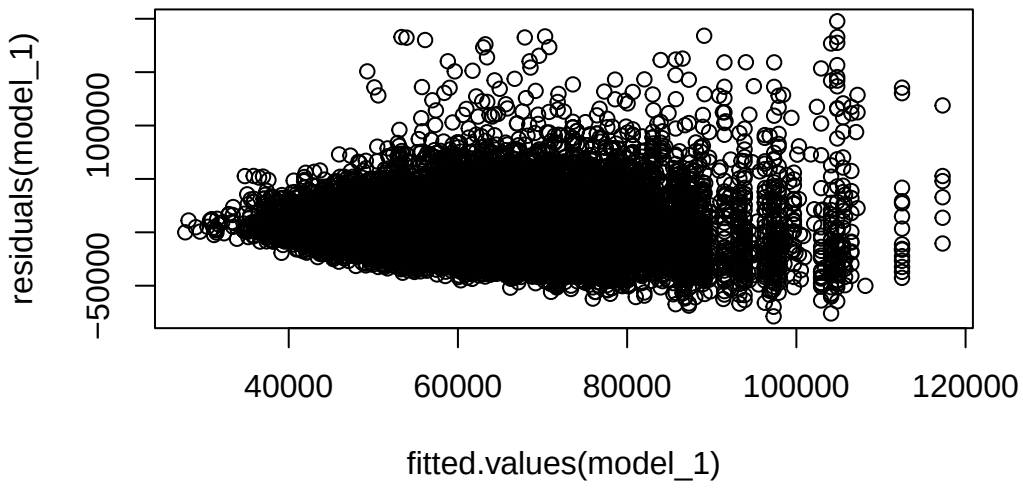
Residual plots and checks for the model's assumptions are shown. Outliers, influential points, or multicollinearity (VIF) are examined. The report says what the checks found, even if nothing was wrong.

Model 1

Testing for Homoscedasticity

One of the assumptions for linear regression is that we have equal variance from the model. This means if we were to plot fitted values versus residuals, we should get equal spread from zero. However, after plotting that, we can see that as our predicted value gets larger, so does the variance. Thus we have evidence against our assumption of equal variance. We can confirm this by running Brown Forsythe test, which gives us a p-value of `2.2e-16`, which suggests it is unlikely our data has equal variance. This suggests a transformation of our outcome variable may be useful.

```
plot(fitted.values(model_1), residuals(model_1))
```




```
leveneTest(residuals(model_1) ~ (fitted.values(model_1) > median(fitted.values(model_1))))
```

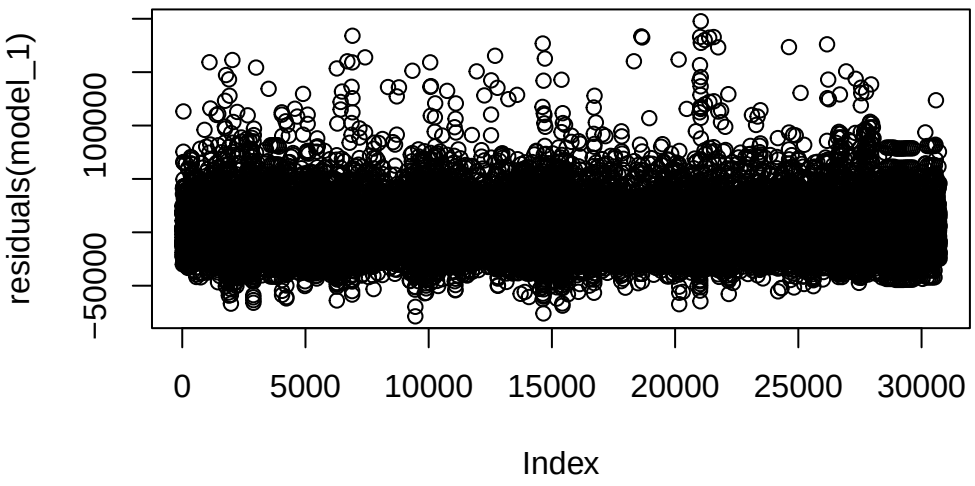
```
Warning in leveneTest.default(y = y, group = group, ...): group coerced to
factor.

Levene's Test for Homogeneity of Variance (center = median)
         Df F value    Pr(>F)
group     1  1390.7 < 2.2e-16 ***
      30731
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Uncorrelated Observations

To test if our observations are independent, we will plot residuals versus index. After doing that, we have equal spread. All of our data comes from the same year, and we don't have observations of the same thing over a few years, so we shouldn't have to worry about autocorrelation.

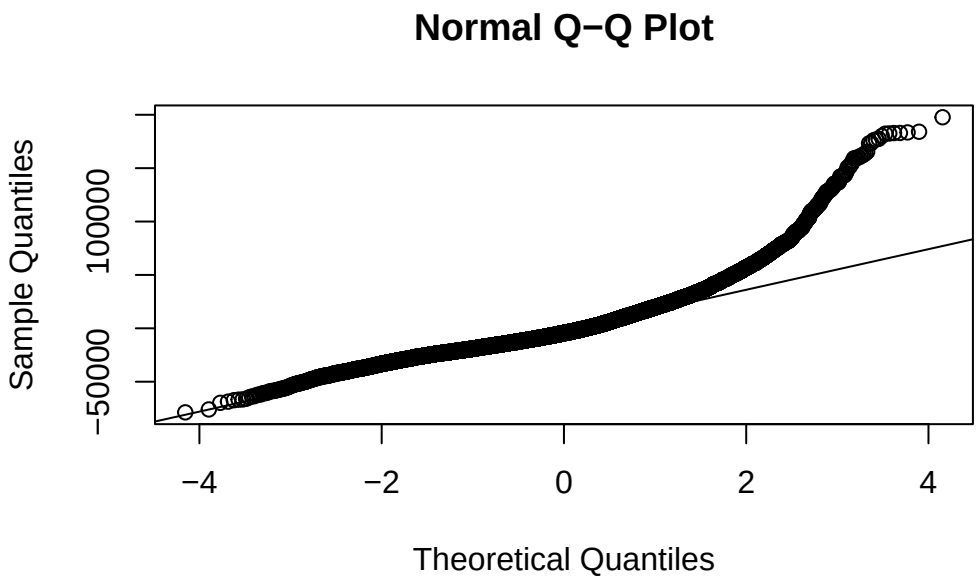
```
plot(residuals(model_1))
```



Normality

To test normality, we will begin by plotting the qqplot. If our data is normally distributed, then our data should fit the qqline very well. However, we can see out data does not, and with the size of our dataset, it is extremely unlikely our residuals are normally distributed.

```
qqnorm(summary(model_1)$residuals)
qqline(summary(model_1)$residuals)
```



Multicollinearity

The last thing we want to look for is collinearity. When predictors are correlated alot, variance increases. We can measure this by using variance inflation factor.

```
print(vif(model_1))
```

```
                                GVIF Df GVIF^(1/(2*Df))
X                           5.733439  1        2.394460
SCH_DEG                     9.573331  1        3.094080
MAIN.x                     12.712348  1        3.565438
NUMBRANCH                 560.642596  9        1.421362
PREDDEG                     9.472695  1        3.077774
HIGHDEG                     1.891320  2        1.172712
ST_FIPS                     5.878733  1        2.424610
REGION                     52.152871  8        1.280353
ADM_RATE_ALL                2.999625  1        1.731943
PCIP12                      1.202250  1        1.096472
PCIP24                      1.349764  1        1.161793
PCIP51                      1.773569  1        1.331754
PCIP52                      1.616864  1        1.271559
UGDS                       43.026407  1        6.559452
```

UG25ABV	7.556947	1	2.748990
UGDS_WHITE	9.686033	1	3.112239
UGDS_BLACK	5.861150	1	2.420981
UGDS_HISP	5.771149	1	2.402322
PPTUG_EF	6.348449	1	2.519613
COSTT4_A	53.338668	1	7.303333
TUITIONFEE_IN	75.901094	1	8.712123
TUITIONFEE_OUT	10.768165	1	3.281488
INEXPFTE	3.180183	1	1.783307
AVGFACSAL	6.455137	1	2.540696
PFTFAC	1.383025	1	1.176021
C150_4	36.391850	1	6.032566
PFTFTUG1_EF	6.530200	1	2.555426
D150_4	24.921125	1	4.992106
C150_4_WHITE	18.325741	1	4.280857
C150_4_BLACK	5.362728	1	2.315756
C150_4_HISP	5.362970	1	2.315809
DEATH_YR8_RT	2.106409	1	1.451347
COMP_ORIG_YR8_RT	15.994972	1	3.999371
COMP_4YR_TRANS_YR8_RT	7.946065	1	2.818877
PELL_COMP_4YR_TRANS_YR8_RT	6.209189	1	2.491824
PELL_WDRAW_ORIG_YR8_RT	9.602988	1	3.098869
NOPELL_WDRAW_ORIG_YR8_RT	8.044658	1	2.836311
NOT1STGEN_COMP_4YR_TRANS_YR2_RT	1.317140	1	1.147667
FIRSTGEN_COMP_ORIG_YR2_RT	2.623652	1	1.619769
TUITFTE.1	7.538311	1	2.745598
OVERALL_YR2_N	41.690964	1	6.456854
DEBT_MDN	2.647684	1	1.627171
DEBT_ALL_STGP_ANY_MEAN	1.096896	1	1.047328
BBRR1_FED_UG_DFLT	1.531274	1	1.237447
DISTANCE	1.106233	1	1.051776
CONTROL.y	13.570375	2	1.919323
SATVRMID	26.055127	1	5.104422
SATMTMID	27.848083	1	5.277128

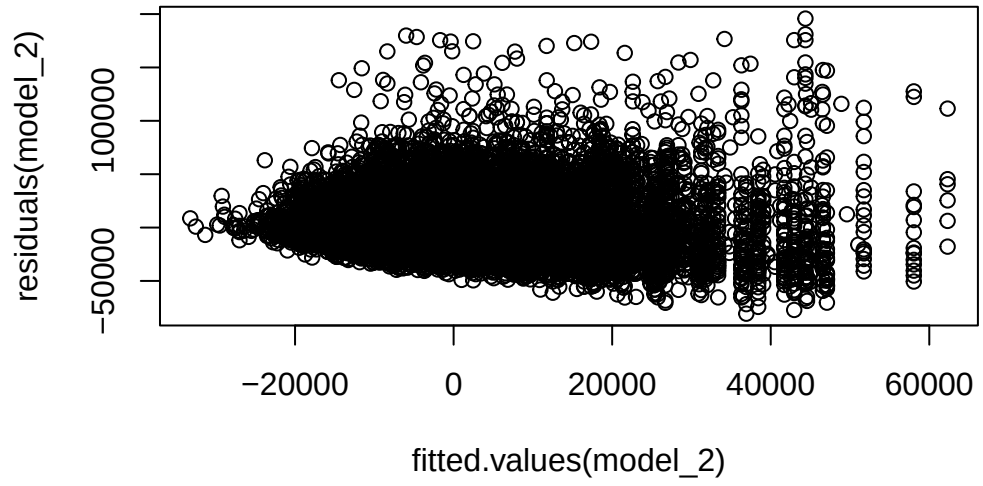
Model 2

Since the assumptions for linear regression are the same for each model, we will run the same tests

Testing for Homoscedasticity

One of the assumptions for linear regression is that we have equal variance from the model. This means if we were to plot fitted values versus residuals, we should get equal spread from zero. However, after plotting that, we can see that as our predicted value gets larger, so does the variance. Thus we have evidence against our assumption of equal variance. We can confirm this by running Brown Forsythe test, which gives us a p-value of `2.2e-16`, which suggests it is unlikely our data has equal variance. This suggests a transformation of our outcome variable may be useful, such as box cox or log transformation.

```
plot(fitted.values(model_2), residuals(model_2))
```




```
leveneTest(residuals(model_2) ~ (fitted.values(model_2) > median(fitted.values(model_2))))
```

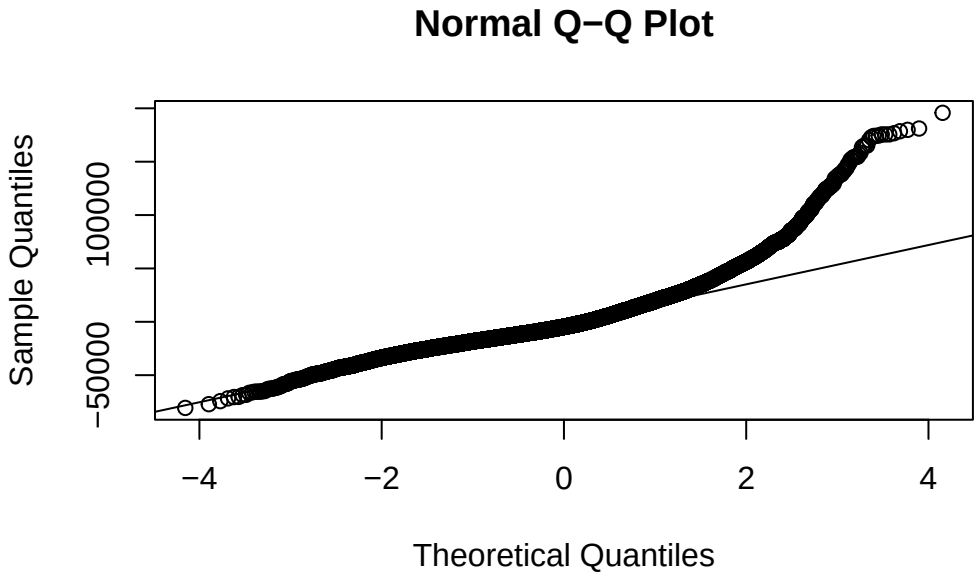
```
Warning in leveneTest.default(y = y, group = group, ...): group coerced to
factor.

Levene's Test for Homogeneity of Variance (center = median)
         Df F value    Pr(>F)
group     1  1284.1 < 2.2e-16 ***
      30731
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Testing for normality

One of the assumptions in regression is normally distributed errors. We will test this by plotting the qqline and qqplot. If our residuals are normally distributed, they should follow the line. If non-normal, they will deviate from it. And since our data-set is so large, it is very unlikely this many points sampled from a normal distribution are this far off normally distributed. This again suggests some transformation of outcome variables, like box cox.

```
qqnorm(summary(model_2)$residuals)
qqline(summary(model_2)$residuals)
```



Multicollinearity

To test for multicollinearity, we will calculate the vif. Using the rough calculation of a variance inflation factor of 10 or larger suggests significant correlation between predictors, we see that happens in a few locations; `C150_4_WHITE` polynomial terms, the 2 different SAT scores having a vif score of around 20 each, though their interaction term only had a vif of 3, and we will take note of `DEATH_YR8_RT` having a vif of `9.978195` and `DEATH_YR8_RT:PELL_COMP_4YR_TRANS_YR8_RT` having a vif of `9.093520`.

```
print(vif(model_2))
```

```
there are higher-order terms (interactions) in this model
consider setting type = 'predictor'; see ?vif
```

	GVIF	Df	GVIF^(1/(2*Df))
MAIN.x	2.357275	1	1.535342
ADM_RATE_ALL	2.674420	1	1.635366
poly(C150_4_WHITE, 2)	12.406537	2	1.876777
PCIP51	1.943209	1	1.393990
PCIP52	1.380699	1	1.175032
poly(UGDS_WHITE, 2)	2.144015	2	1.210060
INEXPFTE	2.897050	1	1.702072
AVGFACSAL	3.562731	1	1.887520
DEBT_ALL_STGP_ANY_MEAN	1.106384	1	1.051848
DEBT_MDN	1.530434	1	1.237107
COMP_ORIG_YR8_RT	7.988913	1	2.826467
DEATH_YR8_RT	1.533700	1	1.238427
PELL_COMP_4YR_TRANS_YR8_RT	1.832299	1	1.353625
NOPELL_WDRAW_ORIG_YR8_RT	3.229569	1	1.797100
DISTANCE	1.058903	1	1.029030
SATVRMID	18.452560	1	4.295644
SATMTMID	20.544702	1	4.532626
PCIP51:PCIP52	1.745561	1	1.321197
AVGFACSAL:DEBT_ALL_STGP_ANY_MEAN	1.164242	1	1.079001
AVGFACSAL:DEBT_MDN	1.194235	1	1.092811
DEBT_ALL_STGP_ANY_MEAN:DEBT_MDN	1.068054	1	1.033467
DEATH_YR8_RT:PELL_COMP_4YR_TRANS_YR8_RT	1.258186	1	1.121689
SATVRMID:SATMTMID	3.289327	1	1.813650

Model 3

Multicollinearity

We know that by selection of our variables, none of our predictors are perfectly multicollinear, as we removed all of the perfectly or (nearly perfectly) multicollinear predictors in a prior part of the project. However, there are several variables which do have inflated VIF measurements, and thus are under suspicion for multicollinearity. Specifically, our in-state tuition, the type of school (public, private nonprofit, or private for-profit), the interaction terms between those two, and the interaction term between school type and admission rate. To some degree, inflated VIF for interaction terms is expected, however, these values are extremely high. Between these measurements and the associated p-values in the model fit, we have reason to remove some, if not most, of these in a revised model. However, as the purpose of this comparison is to see what makes a good model, we continue examining this model rather than a reduced model.

```
vif(model_3)
```

```
there are higher-order terms (interactions) in this model
consider setting type = 'predictor'; see ?vif
```

```
                                GVIF Df GVIF^(1/(2*Df))
TUITIONFEE_IN              49.393192  1        7.028029
CONTROL.x                  42.868320  2        2.558787
ADM_RATE_ALL                6.156807  1        2.481291
PREDDEG                     1.045150  1        1.022326
HIGHDEG                     1.126859  2        1.030309
C150_4                      2.103006  1        1.450174
DEBT_MDN                    1.739878  1        1.319045
TUITIONFEE_IN:CONTROL.x    91.285466  2        3.091010
TUITIONFEE_IN:ADM_RATE_ALL 19.477271  1        4.413306
CONTROL.x:ADM_RATE_ALL     53.219646  2        2.700959
ADM_RATE_ALL:C150_4         3.154716  1        1.776152
```

Testing for Homoscedasticity

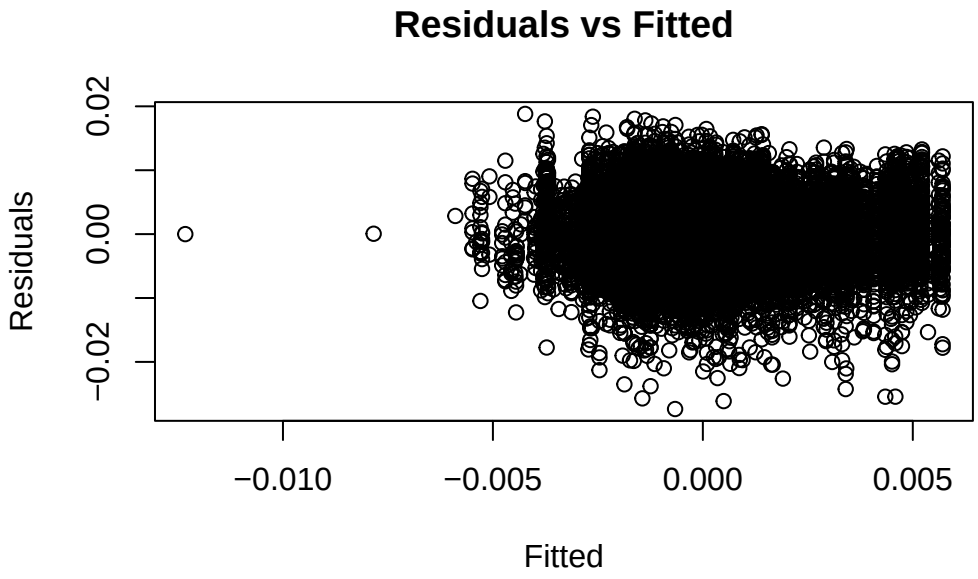
Given our fitted vs residuals plot, we see some indications of heteroskedasticity, in particular, the tendency for residuals to have a slightly more positive value on average, with those having negative values being less dense. Additionally, performing the Brown-Forsythe test gives us a p-value of $7.603 \cdot 10^{-7}$, so Brown-Forsythe concludes that it is unlikely our data has equal variance.

```
leveneTest(residuals(model_3) ~ (fitted.values(model_3) > median(fitted.values(model_3))))
```

```
Warning in leveneTest.default(y = y, group = group, ...): group coerced to
factor.
```

```
Levene's Test for Homogeneity of Variance (center = median)
         Df F value    Pr(>F)
group     1  24.466 7.603e-07 ***
      30731
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

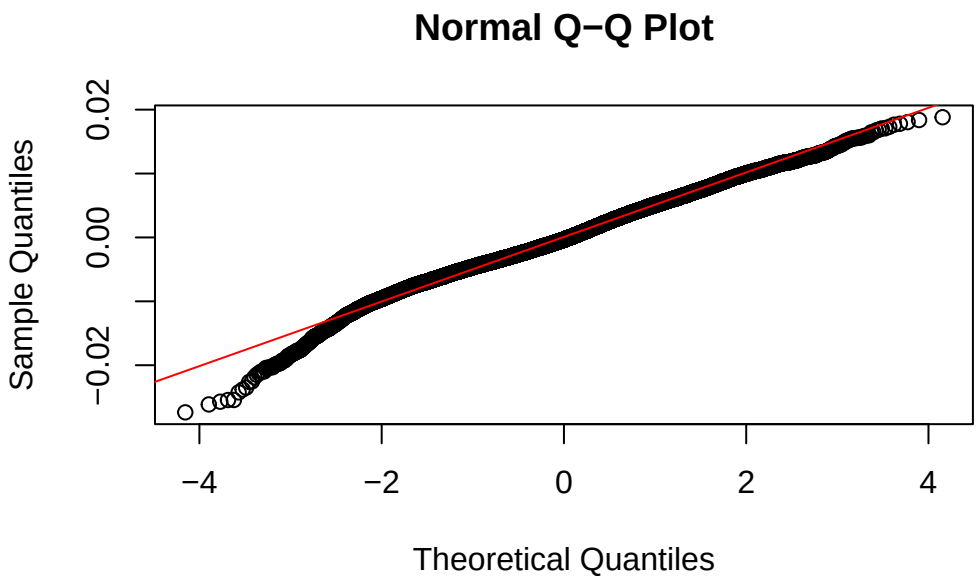
```r
plot(fitted.values(model_3), residuals(model_3), main = "Residuals vs Fitted", xlab = "Fitte
```



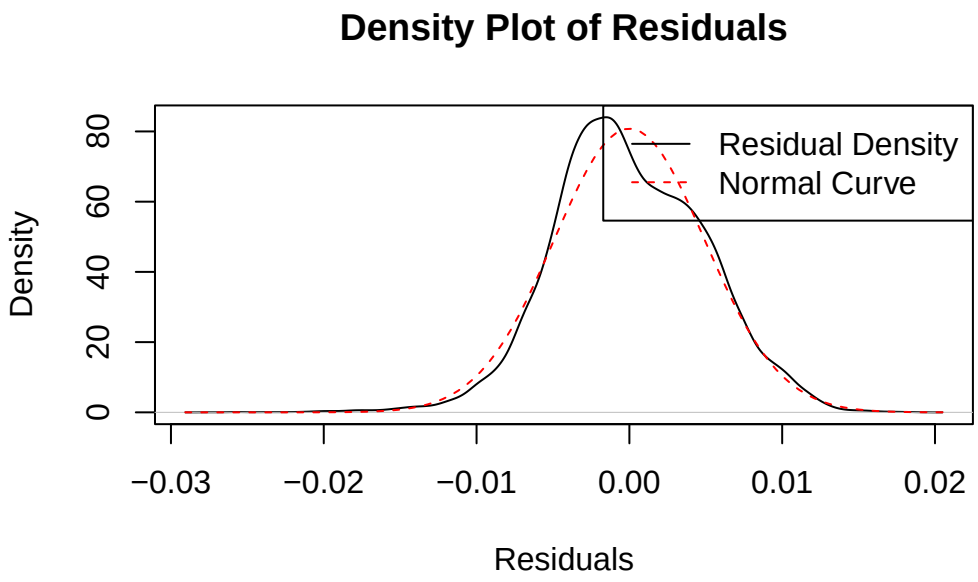
Residuals

Our residuals roughly follow a normal distribution, as established both by our residuals density plot, and our QQ-plot. Our QQ-plot does have some poorly behaved data points at the lower end, however, given the nature of QQ-plots, this is not particularly of concern. As further note for normality of residuals, we did perform a box-cox transform on this model, so the data should be especially well behaved.

```r
qqnorm(summary(model_3)$residuals)
qqline(summary(model_3)$residuals, col = "red")
```




```
plot(density(residuals(model_3)), main = "Density Plot of Residuals", xlab = "Residuals")
curve(dnorm(x, mean = mean(residuals(model_3)), sd = sd(residuals(model_3))),
      col = "red", lty = 2, add = TRUE)
legend("topright", legend = c("Residual Density", "Normal Curve"),
       col = c("black", "red"), lty = c(1, 2))
```



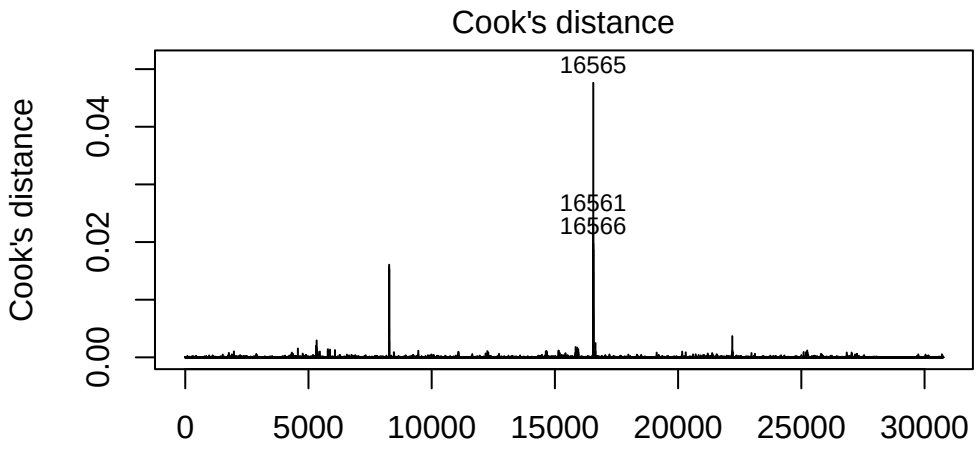
Autocorrelation of Data

Autocorrelation of data is not a concern here, as this data was all collected within a single year. Furthermore, this data is all gathered from disparate institutions, so this further establishes independence from a design perspective.

Outliers and Influential Points

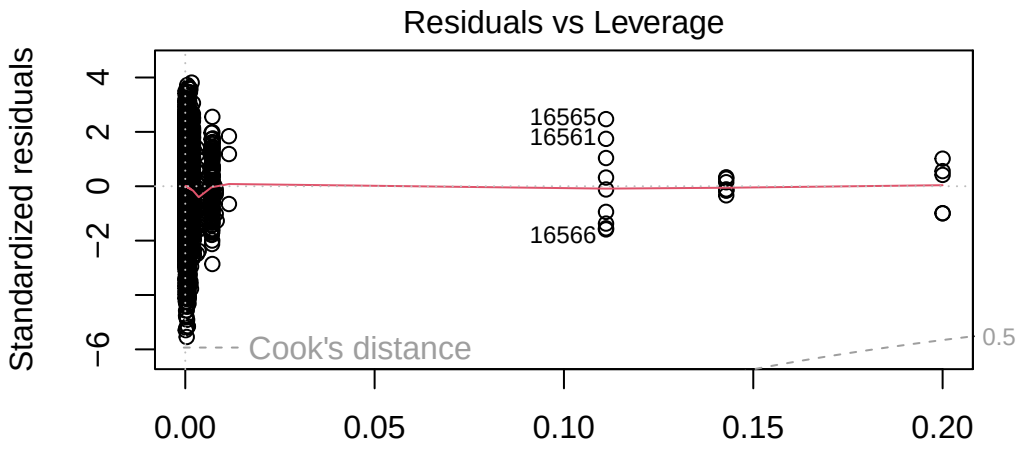
For outliers and influential points, we do find that there are several points with relatively high Cook's Distance, and some with relatively high Cook's Distance with a moderately high leverage. However, given the size of our data-set, we do not believe these are of particular concern. It would, to some extent, be even more odd if there were no such outliers. It is worth noting, however, that some of these more extreme Cook Distance measures are mollified by the Box-Cox transformation we performed earlier. However, we certainly have no reason or motivation to remove these outliers from the existing data.

```
plot(model_3, which = 4)
```




```
plot(model_3, which = 5)
```

```
Warning: not plotting observations with leverage one:
  22736
```



_MDN_4YR ~ (TUITIONFEE_IN + CONTROL.x + ADM_RATE_ALL)^2 + P

4. Model selection and justification (10 pts)

Candidate models are compared with a stated criterion (adjusted R^2, AIC, BIC, C_p, or an F test). The final model is chosen and the choice is justified in words. Note: STAT 540 groups: deeper selection justification In addition to the above, a 540 group compares at least three candidate models using at least two different criteria. Where the criteria disagree, the report explains why and says what the group traded off (fit against parsimony, or prediction against interpretation). Award 10 only if this is done. This is part of the criterion, not extra points.

Model Selection

The two benchmarks we will use to measure how well our models perform are `R^2` and `AIC`.

R squared

```
summary(model_1)$r.sq
```

```
[1] 0.2051042
```

```r
summary(model_2)$r.sq
```

```
[1] 0.2139218
```

```r
summary(model_3)$r.sq
```

```
[1] 0.1152715
```

Looking at our models, we see model 1, the full linear model with no interaction or polynomial terms explains `0.2051042` proportion of variation from mean, model 2 explains slightly more variation from the mean at `0.2139218`, and model 3 explains `0.1152715` proportion of the variation from the mean. This tells us looking at polynomial and interaction terms does explain a meaningful amount of variation from the mean. This also suggests that model 3 cut useful terms.

AIC

```r
extractAIC(model_1)[2]
```

```
[1] 616681.8
```

```r
extractAIC(model_2)[2]
```

```
[1] 616258.9
```

```r
extractAIC(model_3)[2]
```

```
[1] -326418.7
```

Looking at AIC, which is a measure of how much information is lost when reducing models with a penalty for more complex models we get that the best model is model 3, with models 1 and 2 about the same as each other with model 2 slighly smaller than model 1. This suggests that model 3 explains much of the data with way less predictors/interaction/polynomial terms.

Selected Model and Justification

After looking at these measures, we conclude that model 2 is in general a better model. It has the lower AIC (with model 3 having a negative AIC due to lack of predictive power), and it also has the highest R^2 value, by nearly a full percentage point.

However, as model 2 failed several key assumptions of a linear model, it does need to be slightly reworked. In particular, we need to perform a Box-Cox transformation to fix several key assumptions with the residuals.

5. Integration and reproducibility (10 pts)

The report builds on Milestone 2 and responds to its feedback. The submission is a rendered PDF named Group_XX_Milestone3.pdf, with code visible, a seed set, and sessionInfo() present.

For Diagnostics and Reproducibility:

```
sessionInfo()
```

```
R version 4.4.3 (2025-02-28 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default


locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/Denver
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
```

```
 [1] erikmisc_0.2.91 MASS_7.3-65     car_3.1-3       carData_3.0-5
 [5] lubridate_1.9.4 forcats_1.0.1   stringr_1.6.0   dplyr_1.1.4
 [9] purrr_1.2.1     readr_2.1.6     tidyr_1.3.2     tibble_3.3.1
[13] ggplot2_4.0.1   tidyverse_2.0.0

loaded via a namespace (and not attached):
  [1] fs_2.0.1                bitops_1.0-9            insight_1.6.0
  [4] RColorBrewer_1.1-3      doParallel_1.0.17       numDeriv_2016.8-1.1
  [7] tools_4.4.3             backports_1.5.0         sjlabelled_1.2.0
 [10] R6_2.6.1                mgcv_1.9-4              nortest_1.0-4
 [13] jomo_2.7-6              withr_3.0.2             gridExtra_2.3
 [16] quantreg_6.1            cli_3.6.5               textshaping_1.0.4
 [19] officer_0.7.7           sandwich_3.1-1          cAIC4_1.1
 [22] mvtnorm_1.3-3           S7_0.2.1                robustbase_0.99-7
 [25] askpass_1.2.1           systemfonts_1.3.1       yulab.utils_0.2.5
 [28] foreign_0.8-90          svglite_2.2.2           qqplotr_0.0.7
 [31] DHARMa_0.5.0            parallelly_1.46.1       labelled_2.16.0
 [34] readxl_1.6.0            pwr_1.3-0               rstudioapi_0.18.0
 [37] visNetwork_2.1.4        generics_0.1.4          gridGraphics_0.5-1
 [40] shape_1.4.6.1           sjPlot_2.9.0            zip_2.3.3
 [43] leaps_3.2               Matrix_1.7-4            ggbeeswarm_0.7.3
 [46] abind_1.4-8             lifecycle_1.0.5         multcomp_1.4-29
 [49] yaml_2.3.12             gvlma_1.0.0.3           snakecase_0.11.1
 [52] ggpmisc_1.0.0           recipes_1.3.1           grid_4.4.3
 [55] crayon_1.5.3            mitml_0.4-5             lattice_0.22-7
 [58] haven_2.5.5             cowplot_1.2.0           pillar_1.11.1
 [61] knitr_1.51              log4r_0.5.0             boot_1.3-32
 [64] estimability_1.5.1      future.apply_1.20.1     codetools_0.2-20
 [67] pan_2.0                 glue_1.8.0              fontLiberation_0.1.0
 [70] data.table_1.18.0       vctrs_0.7.0             png_0.1-9
 [73] Rdpack_2.6.4            cellranger_1.1.0        gtable_0.3.6
 [76] twosamples_2.0.1        ggpp_0.6.1              gower_1.0.2
 [79] xfun_0.56               rbibutils_2.4           prodlim_2025.04.28
 [82] pracma_2.4.6            coda_0.19-4.1           reformulas_0.4.3.1
 [85] survival_3.8-6          randomForestSRC_3.9.0   timeDate_4051.111
 [88] iterators_1.0.14        pbmcapply_1.5.1         tinytex_0.58
 [91] finalfit_1.1.0          hardhat_1.4.2           lava_1.8.2
 [94] DiagrammeR_1.0.12       TH.data_1.1-5           ROCR_1.0-12
 [97] ipred_0.9-15            nlme_3.1-168            tsne_0.2-0
[100] fontquiver_0.2.1        data.tree_1.2.0         vipor_0.4.7
[103] otel_0.2.0              rpart_4.1.24            colorspace_2.1-2
[106] Hmisc_5.3-0             nnet_7.3-20             tidyselect_1.2.1
[109] emmeans_2.0.1           compiler_4.4.3          glmnet_4.1-10
```

```
[112] htmlTable_2.5.0        flextable_0.10.3       mice_3.19.0
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[127] pkgconfig_2.0.3        base64enc_0.1-3        lme4_1.1-38
[130] umap_0.2.10.0          fastmap_1.2.0          rlang_1.1.7
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[136] zoo_1.8-15             jsonlite_2.0.0         ModelMetrics_1.2.2.2
[139] magrittr_2.0.4         polynom_1.4-1          kableExtra_1.4.0
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[169] MatrixModels_0.5-4     openssl_2.3.4          future_1.69.0
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[178] class_7.3-23           ragg_1.5.0             lmerTest_3.2-0
[181] beeswarm_0.4.0         ellipse_0.5.0          cluster_2.1.8.1
[184] corrplot_0.95          timechange_0.3.0       globals_0.18.0
[187] caret_7.0-1
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